BITU 2913 – WORKSHOP 1

BLOOD BANK MANAGEMENT SYSTEM

Presented by Tan Xin Rou (B032410560)
Supervisor: En. Amirul Ramzani Radzid
Evaluator: Ts. Dr. Raja Rina binti Raja Ikram

Introduction

In today's world, many organizations in Malaysia still rely on **paper-based systems** that cause **unreadable records, data loss, security issues, and slow donor eligibility verification**. These manual methods make it difficult to **track donor information, blood group availability, and blood expiry dates**, leading to **inefficiency, human errors, and possible blood shortages or wastage**.

This project introduces a **Blood Bank Management System** that **digitalizes donor and donation records**, enabling **accurate, secure, and fast data retrieval**. Overall, the system **reduces paperwork, improves efficiency, and ensures the right blood is available when needed**.

Problem Statement



Difficulty in retrieving donation information

- forces staff to spend excessive time searching through manual records.



Inefficiency in manual documentation

- makes recording donor and donation details slow and prone to errors.



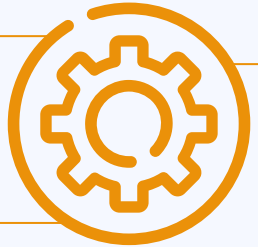
Risk of data loss from poor organization and insecure storage

- can lead to missing important information and potential harm.

Objectives

To handle donors and staff details efficiently

To maintain all records in a centralized database



To improve data accuracy and reduce human error

Scope



Staff/Admin

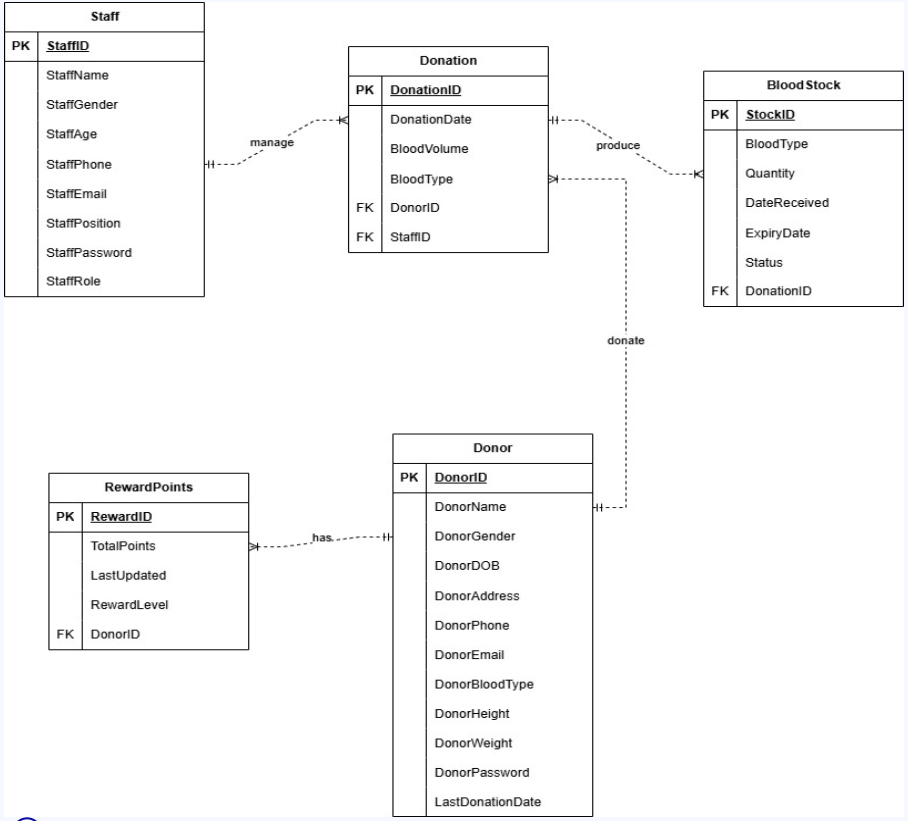
Manage and maintain donor, staff, and blood stock records, including creating, updating, deleting, retrieving information, and generating reports.



Donor

View their personal information, track donation history, and monitor their points and rewards.

Entity Relationship Diagram



Functionalities

For Admin

- Create, read, update, delete staff details
- View staff records, staff related reports

For Staff

- Create, read, update, delete donor details, blood stock details, donation details
- View recent donation records, blood donation report, account information

For Donor

- View self donation record, reward points, account information

Extra Features

Exporting reports to text file

- After staff / admin view a report, system will ask if the user wants to export reports to text file.
- To allow staff or administrators to save, share, and review report data easily for record-keeping, analysis, and documentation purposes.

View Donor's Reward Point

- The system will automatically add 10 reward points to a donor's account after each donation.
- To reward donor's contributions and to increase donor engagement and loyalty.

Forget Password

- To allow users to reset password if they forget it.
- Ask for id and phone number for verification.

SYSTEM DEMONSTRATION

Limitations

- The system is text-based, runs locally on a single device, and lacks network or multi-hospital support.
- Reporting is basic, providing only simple staff and donation reports without advanced analytics or trend predictions.
- Passwords and sensitive donor data are stored without encryption.

Future Improvement

- Use GUI or web-based platform with network support
- Reporting includes analytics, such as blood type demand prediction, donation trends, and cross-branch reporting.
- Passwords and sensitive data should be encrypted, with multi-factor authentication for security.
- Mobile support with SMS notifications.

THANK YOU